Course Syllabus

1. General Information

Name of the Faculty	Dr. Abdullah Al Nuaim
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Office Location	Room No: 2034
Office hours	Modnay 11:30 AM – 1:00 PM Tuesday 11:30 AM – 1:00 PM **Office Hours are held virtually via MS Teams**
College/department	School of Business/ Management Information Systems
Course Title /Code	Project I / 0677303
Credit hours	2

Course Description:

The purpose of this course is to provide students with an opportunity to apply the knowledge they have acquired, their intellectual abilities and practical skills to solve real-life business problems. The students are required to carry out a research or a systems development project. This part of the project will comprise the initial activities pertaining to the problem definition and solution design.

Course Main Objective

The purpose of this course is to integrate the various components of knowledge, skills, and competencies acquired within the program and provide the students with an opportunity to practice their knowledge.

Course Learning Outcomes

- **Knowledge**
  - Recognize the economic, social, and environmental issues involved in the design and development of business solutions or research.
  - Describe the organizational performance/ efficiency issues to be undertaken.
  - Describe the activities involved in the system development lifecycle.
  - Outline the role of advanced technologies in achieving business objectives.
- **Skills**
  - Analyze the requirements of a project.

- Design a solution to a given business problem.
- Demonstrate effective oral and written communication skills.
- Demonstrate innovation in proposing and developing business solutions.

- **Values**
  - Demonstrate effective time management and organization skills while working with individuals, teams, and large groups.
  - Demonstrate awareness of ethics, values and standards involved in design and development of business solutions.
  - Demonstrate the ability to take initiatives in acquiring knowledge and skills, as well as follow a self-committed plan.

Learning Resources:

- Required Textbook:
  - Systems Analysis and Design, Kendall, K.E., Kendall, J.E. 10th Edition (2018), Prentice-Hall.
  - Systems Analysis and Design, Alan Dennis, Barbara Haley Wixom, Roberta M. Roth, 7th Edition (2018), Wiley.
- Essential References: Instructor handouts
- Electronic Materials: Links will be distributed to students during class

Course Schedule

Week	Topic	Notes
1	Course outline & Introduction	
2	Defining the problem	
3	Development of the project plan using a Gantt chart	
4	Project feasibility study	
5	Information gathering	
6	Requirements analysis and prioritization	
7	Building and evaluating the solution prototypes	
8		
9	Solution design	
10		
11	Evaluation of the final design	
12	Presentation and oral examination	Project due
13		

Assessment Plan

Grading Scheme			
Assessment	Assessment task	Week due	Percentage of Total Assessment Score
1	Project-I report		75 %
2	Presentation and oral examination		25%
	Total		**100%**

NOTE

Academic integrity: Rules listed in the Calendar will be strictly enforced.

Proposed project idea (3 pages maximum)

In the first phase of your project activities, you need to propose an idea. You are asked as a team to prepare a proposed project idea report that includes the following details:

Information System Title

Develop a title that is relevant to the work to be done in the project and that fits in one line.

Information System Background

Write a summary of the business for which the Information system is to be developed, or the new business that will be created with this information system (e.g., online business, mobile app, digital platform, etc.). The details should include: a description of the business, history of the business, location, market, and target customers.

Information System statement

Explain in one paragraph what the proposed Information system is about.

Duration of the Project

Develop a table to show the expected duration of the project. This should show all phases as listed in the course syllabus, the tasks identified for each phase, and the duration for each task. This can be updated later on when you receive the full project specification document.

Project Team Members' Responsibilities

This should include: the team member's name and responsibilities. This can be updated later on when you receive the full project specification document.

Supplementary notes:

No mark is given for this part. You need to obtain approval for your proposal from your instructor before starting other phases of the project.

The work should be the group's own work and any attempt to copy from a previous work or other group(s) work will be considered as plagiarism.

Deadline for submission: end of the second week (see attached calendar)

Group size: 5 students only

Analysis and Design Phase specifications document

Report chapter	Task	Details
Chapter 1: Preliminary Investigation	1.1 Problem Statement	Write one paragraph explain why this system is needed and what business problem it is solving
	1.2 Solution	Write one paragraph explain how the proposed system will solve the above business problem
	1.3 Scope of Work / Project Scope	What system does / does not cover (boundaries)
	1.4 Feasibility study:	Discuss all 3 types of feasibility
	1.5 Functional Requirements	List main functionalities of the proposed system
	1.6 List of staff's functions	List what the major roles in the proposed system
	1.7 Data Collection Techniques	What techniques were used to collect the data for your study (at least two techniques)
	1.8 System Requirements:	List all the Hardware, software platform and Operating systems types needed for this system
	1.9 Technologies	This includes tools used in all stages of the SDLC
	1.10 Project Timeline	Time estimates for each stages of project with phases and tasks – distribution of allocated weeks /days (end date is given)
Chapter 2: System analysis	2.1 Context Diagram	Draw Context Diagram, and provide the description of the context diagram, Describe The Interaction Between The System And Its External Entities In The Diagram
	2.2 Functional Decomposition Diagram	Draw FDD diagram

Report chapter	Task	Details
	2.3 Data Flow Diagram	(LEVEL O) Describes shortly what each component is about
Chapter 3: System Design	3.1 input and output forms design	Design screen forms and output (reports) of your system
	3.2 Prototype design	Provide a prototype showing a basic user interface design covering the major functions of the system.
	3.3 Entity Relationship Diagram	Draw an ER diagram for your system
	3.4 Normalization	Perform normalization to 3rd normal form
	3.5 Logical database design	Show the logical design of your proposed system (Relational database design)
	3.6 Physical database design	Here, provide the implementation-level details of all the tables, fields and relationships.

What expected from you next semester in Project II

Specifications for the Implementation Phase

This document covers the details of activities to be covered while carrying out the development, testing, documentation, and deployment of the system and lists the topics to be covered in the respective chapters.

Important note: You are not required to submit an individual report for the implementation phase. Instead, the chapters given in the following will be included in the final project report that needs to be submitted as a whole by the 14th week of the semester. However, if you would like to get the supervisor's feedback on your work, you may submit it in parts.

Implementation Phase Deliverables and Marks Allocation

Chapter	Specific tasks to be documented	Details
Chapter 4: Development	4.1 Tools/ technologies selected for implementation	Discuss how all tools/ technologies selected for the implementation of this system have been employed. Justify the selection where applicable.
	4.2 Web system navigation design	Present the overall navigation design in the form of a figure and discuss it. Refer to the provided template.
	4.3 User interface design	Provide the details of user interface design covering major functions of the system. Highlight the measures taken to

		increase the usability of the user interface (e.g., form validation, using regular expressions, etc.).
	4.4 Database connectivity	Discuss in detail the mechanism used to integrate the user interface with the backend database.
Chapter 5: Testing and conversion	5.1 Test plan	Provide the details of the test plan to carry out unit, system and integration testing of the system. See supplementary notes at the end for more details.
	5.2 Unit testing	Provide details of the unit testing along with synopsis of selected results. See supplementary notes at the end for more details.
	5.3 Integration testing	Provide details of the integration testing along with synopsis of selected results. See supplementary notes at the end for more details.
	5.4 System testing	Provide details of the system testing along with synopsis of selected results. See supplementary notes at the end for more details.
	5.5 Conversion plan	Provide details of the plan for conversion of the old information system to the new one. Specify the conversion strategy to be used and justify it.
Chapter 6: System documentation	6.1 User manual	Create and provide details of the user manual. This should cover the assistance for all major roles within the system.
	6.2 Technical support manuals	Create and provide details of the manuals for system administrators, maintenance personnel.
	6.3 Online help and FAQs	Create and provide details of the online help and FAQs.

Supplementary Notes

5.1 Test plan

Here, the focus should be on the following:

(1) Requirements for test

It should focus on what will be tested. List those items (use cases, functional requirements, non-functional requirements) that have been identified as targets for testing.

(2) Test strategy

It presents the recommended approach to the testing of the software applications. The previous section described what will be tested; this describes how it will be tested. The main considerations for the test strategy are the techniques to be used and the criterion for knowing when the testing is completed.

For a more detailed example of a test plan, visit this link:

http://sce.uhcl.edu/helm/RUP_course_example/courseregistrationproject/artifacts/test/plans/test_ppla_arch.htm

Note, however, that you are not required to cover all the details given in that example.

5.2 Unit testing

Unit testing is the process of testing program components, such as methods or object classes. Individual functions or methods are the simplest type of component. Your tests should be calls to these routines with different input parameters. Proposed details to be included for a unit test are shown in the table below.

Test Case Name	
Purpose	
Component Name	
Component Type (method/ object)	
Test Description (Steps)	
Input Provided	
Expected Outcome	
Actual Outcome	
Status (Pass/Fail)	
Comments	

5.3 Integration testing

Integration testing (or component testing) is carried out after unit testing and it relates to testing of software components, which are made up of several interacting objects. It focuses on showing that the component interface behaves according to its specification. Here, the test cases are not applied to the individual components but rather to the interface of the composite component created by combining these components.

Here, you should be providing details of the black box tests needed to ensure that each component of system is interacting with other components as expected. Refer to the table below. /

Test Case Name	
Purpose	
External Dependencies	
Test Description (Steps)	
Expected Results	
Actual Results	
Status (Pass/Fail)	
Comments	

5.4 System testing

System testing during development involves integrating components to create a version of the system and then testing the integrated system. Refer to the table below.

Test Case Name	
Purpose	
Test Description (Steps)	
Expected Results	
Actual Results	
Status (Pass/Fail)	
Remarks	